Tutorial – 10
Solutions

$\boxed{\epsilon_{r_1} = 2, \quad \epsilon_{r_2} = 1.5 = \frac{3}{2}}$ (given)

05/03/2016

1 $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$

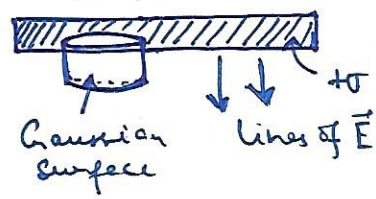
(a) $E = \frac{\sigma}{\epsilon_0}$ between plates (in absence of dielectric)

E will polarize the dielectric, $\vec{P}$ will be induced on the dielectric

Question 1 has been done again in last few pages in a more clearer wayafter incorporating some changes

To find D in each slab:

(Top Slab)

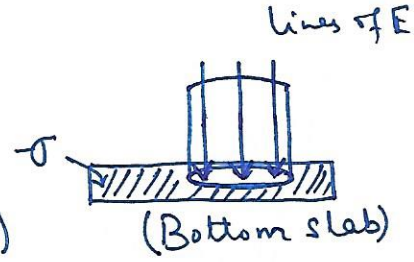


$$\oiint \vec{D} \cdot d\vec{a} = (Q_f)_{enc}$$

Flux of D will only contribute where it encloses free charge - thus only the flux from the circular part of Gaussian cylinder contributes

$D \cdot A = \sigma A \quad \therefore D = \sigma$ for upper slab (1)

For lower slab



$-D \times -A = \sigma A$

$\therefore \sigma = D$ for lower slab. (2)

$D = \sigma$ for each slab and it points downwards. $\boxed{D = 0 \text{ in each conductor}}$

(b) $D = \epsilon E \quad \therefore E_1 = \frac{D_1}{\epsilon_1} = \frac{\sigma}{\epsilon_1} = \frac{\sigma}{\epsilon_0 \epsilon_{r_1}} = \frac{\sigma}{2\epsilon_0}$

$\boxed{\text{Also since } E_{med} = \frac{E_0}{\epsilon_r}, \quad E_1 = \frac{\sigma/\epsilon_0}{2} = \frac{\sigma}{2\epsilon_0}}$

Similarly $E_2 = \frac{D_2}{\epsilon_2} = \frac{\sigma}{\epsilon_0 \epsilon_{r_2}} = \frac{2\sigma}{3\epsilon_0}$

(c) $\vec{P} = \epsilon_0 \chi_e \vec{E}$. But $\epsilon_r = (1 + \chi_e) \Rightarrow \chi_e = (\epsilon_r - 1)$

$\vec{P} = \epsilon_0 (\epsilon_r - 1) \vec{E}$

direction of $\vec{P}$ is from -ve charge to +ve charge $\rightarrow$

$$\begin{cases} P_1 = \epsilon_0 (\epsilon_{r_1} - 1)\, \sigma/2\epsilon_0 = \epsilon_0 (2-1) \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{2} \\ P_2 = \epsilon_0 (\epsilon_{r_2} - 1) \frac{2\sigma}{3\epsilon_0} = \cancel{\epsilon_0} \cdot \frac{1}{\cancel{2}} \cdot \frac{2\sigma}{3\cancel{\epsilon_0}} = \frac{\sigma}{3} \end{cases}$$

(d)

$$V = E_1 a + E_2 a = \frac{\sigma}{2\epsilon_0} a + \frac{2\sigma}{3\epsilon_0} a$$

$$V = \frac{3\sigma a + 4\sigma a}{6\epsilon_0} = \frac{7\sigma a}{6\epsilon_0}$$

(e) location and amount of all bound charge

$\rho_b = 0$ (Polarization is uniform) $[\because \rho_b = -\vec{\nabla}\cdot\vec{P}]$

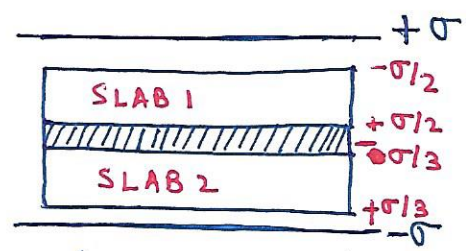
$\sigma_b = \vec{P}\cdot\hat{n}$ We have calculated $P_1 = \frac{\sigma}{2}$ (slab 1), $P_2 = \frac{\sigma}{3}$ (slab 2)

Bound charges and their locations:

Slab (1): Top part $\rightarrow \sigma_b = -\left(\frac{\sigma}{2}\right)$ Bottom part $\rightarrow \sigma_b = +\left(\frac{\sigma}{2}\right)$

Slab (2): Top part $\rightarrow \sigma_b = +\left(\frac{\sigma}{3}\right)$ Bottom part $\rightarrow \sigma_b = \left(-\frac{\sigma}{3}\right)$

(f) Recalculate field in each slab and confirm (b)



(Slabs are continuous at interface)

Field inside slab (1)

Total surface charge on top plate

$$\sigma - \frac{\sigma}{2} = +\frac{\sigma}{2}$$

Total surface charge on bottom plate

$$\frac{\sigma}{2} + \frac{\sigma}{3} - \frac{\sigma}{3} - \sigma = -\frac{\sigma}{2}$$

Field inside slab (1) is $\frac{\sigma}{2\epsilon_0}$

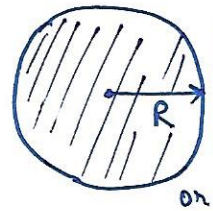
Field inside slab (2)

Total surface charge on top part $= \sigma - \frac{\sigma}{2} + \frac{\sigma}{2} - \frac{\sigma}{3} = \frac{2\sigma}{3}$

" " " " bottom " $= \frac{\sigma}{3} - \sigma = -\frac{2\sigma}{3}$

Field inside slab (2) $= \frac{2\sigma}{3\epsilon_0}$

2 We have to calculate the potential at the center of this dielectric sphere



free charge density $= \rho$

dielectric const $= \epsilon_r$

or permittivity $= \epsilon$

$$V(0) = -\int_{\infty}^{0} \vec{E}\cdot d\vec{r} \quad \text{—— (i)}$$

We have to calculate the electric field inside and outside the sphere. As it is a dielectric sphere, apply Gauss's law in dielectrics:

<u>Region $r<R$</u>

$$\oiint \vec{D}\cdot d\vec{a} = (Q_f)_{enc} \Rightarrow D\cdot 4\pi r^2 = \int \rho\, d\tau' = \rho \int_0^r 4\pi r'^2\, dr'$$

$$= \frac{4}{3}\pi r^3 \rho$$

$$\vec{D}_{in} = \frac{\rho r}{3}\hat{r} \quad \therefore \quad \vec{E}_{in} = \frac{\vec{D}_{in}}{\epsilon}$$

$$= \frac{\rho r}{3\epsilon}\hat{r} \quad [\because \vec{D} = \epsilon\vec{E}]$$

<u>Region $r>R$</u>

$$\oiint \vec{D}\cdot d\vec{a} = (Q_f)_{enc} \Rightarrow D\cdot 4\pi r^2 = \frac{4}{3}\pi R^3 \rho$$

$$\vec{D}_{out} = \frac{\rho R^3}{3r^2}\hat{r}$$

$$\vec{E}_{out} = \frac{\vec{D}}{\epsilon_0} = \frac{\rho R^3}{3\epsilon_0 r^2}\hat{r}$$

According to eqn (i), $V(0) = -\int_{\infty}^{R} \vec{E}_{out}\cdot d\vec{r} - \int_{R}^{0} \vec{E}_{in}\cdot d\vec{r}$

$$= \frac{\rho R^3}{3\epsilon_0}\int_{\infty}^{R}\frac{dr}{r^2} - \frac{\rho}{3\epsilon}\int_{R}^{0} r\,dr = \frac{\rho R^2}{3\epsilon_0} + \frac{\rho R^2}{3\epsilon \times 2}$$

$$V(0) = \frac{\rho R^2}{3\epsilon_0}\left(1 + \frac{1}{2\epsilon_r}\right) \quad \left[\epsilon_r = \frac{\epsilon}{\epsilon_0}\right]$$

<u>Calculation of bound charges</u>

i) $\rho_b = -\vec{\nabla}\cdot\vec{P}$ $\quad \vec{P} = \vec{D} - \epsilon_0\vec{E}$. We need $\vec{D}_{in}$ and $\vec{E}_{in}$ to calculate ρ_b

$$\vec{P} = \left(\frac{\rho r}{3} - \epsilon_0\frac{\rho r}{3\epsilon}\right)\hat{r} \qquad \cancel{\vec{\nabla}\cdot\vec{P} = \frac{1}{r^2}\frac{\partial}{\partial r}\left\{r^2\left(\frac{\rho r}{3} - \frac{\rho r}{3\epsilon}\right)\right\}}$$

$$= \frac{\rho r}{3}\left(1 - \frac{\epsilon_0}{\epsilon}\right)\hat{r} = \frac{\rho r}{3}\left(1 - \frac{1}{\epsilon_r}\right)\hat{r} = \frac{\rho r}{3}\left(\frac{\epsilon_r - 1}{\epsilon_r}\right)$$

In spherical co-ordinates $\vec{\nabla}\cdot\vec{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r)$ if $\vec{v} = \vec{v}(r)$

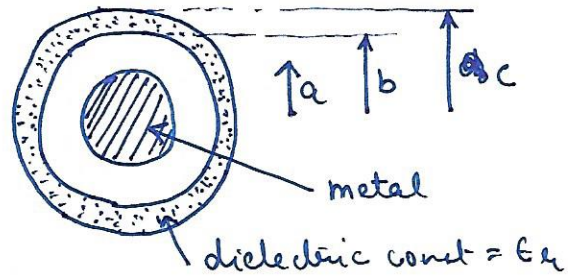
$$P_r = \frac{\rho r}{3}\left(\frac{\epsilon_r - 1}{\epsilon_r}\right) \qquad \rho_b = -\vec{\nabla}\cdot\vec{P} = -\left[\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 \times r)\right]\frac{\rho}{3}\left(\frac{\epsilon_r - 1}{\epsilon_r}\right)$$

$$= -\frac{1}{\cancel{r^2}}\, 3\cancel{r^2}\,\frac{\rho}{3}\left(\frac{\epsilon_r - 1}{\epsilon_r}\right) = \underline{\underline{-\rho\left(\frac{\epsilon_r - 1}{\epsilon_r}\right)}}$$

ii) $\sigma_b = \vec{P}\cdot\hat{n}$. For the sphere $\hat{n} = \hat{r}$

$$\vec{P} = \frac{\rho r}{3}\left(\frac{\epsilon_r - 1}{\epsilon_r}\right)\hat{r} \qquad \therefore \sigma_b = \vec{P}\cdot\hat{n}\Big|_{r=R} = \underline{\underline{\frac{\rho R}{3}\left(\frac{\epsilon_r - 1}{\epsilon_r}\right)}}$$

3. To find the capacitance, let us assume that there is $+Q$ charge in the inner cylinder (radius 'a') and $-Q$ charge on the outer cylinder (inner radius 'c').



Sphere 'a' is metallic. Let's calculate the potential difference between inner and outer cylinder.

$V = V(a) - V(c) = -\int_c^a \vec{E}\cdot d\vec{l}$. To calculate $\vec{E}$, we would use Gauss's law in dielectrics $\oint \vec{D}\cdot d\vec{a} = (Q_f)_{enc}$ since the problem involves dielectric materials.

Assume a Gaussian cylinder of length 'l' and radius 's', where 's' denotes radial distance from the axis

$D \cdot 2\pi s l = Q$. [Of course the Gaussian cylinder has a radius greater than 'a' ($s > a$) so that it could enclose a net free charge Q]

$$\vec{D} = \frac{Q}{2\pi s l}\hat{s}$$

Thus $\vec{E} = \frac{\vec{D}}{\epsilon_0}$ for $a < s < b$ (vacuum) and $\vec{E} = \frac{\vec{D}}{\epsilon}$ $(b < s < c)$

$$\vec{E} = \frac{Q}{2\pi\epsilon_0 s l}(\hat{s}) \quad (a < s < b)$$

$$= \frac{Q}{2\pi\epsilon s l}(\hat{s}) \quad (b < s < c), \quad \epsilon = \text{permittivity} = \epsilon_0\epsilon_r$$

Of course inside the metal $(s < a)$, there is no polarization $\vec{P} = 0$ and electric field, $\vec{E} = 0$, i.e. $\vec{E} = \vec{P} = \vec{D} = 0$

$$V = V(a) - V(c) = -\int_c^a \vec{E}\cdot d\vec{l} = -\int_c^b \vec{E}\cdot d\vec{l} - \int_b^a \vec{E}\cdot d\vec{l} = \int_b^c \vec{E}\cdot d\vec{l} + \int_a^b \vec{E}\cdot d\vec{l}$$

$$= \int_b^c \frac{Q}{2\pi\epsilon l}\frac{ds}{s} + \int_a^b \frac{Q}{2\pi\epsilon_0 l}\frac{ds}{s} = \frac{Q}{2\pi\epsilon_0 l}\left[\ln\left(\frac{b}{a}\right) + \frac{1}{\epsilon_r}\ln\left(\frac{c}{b}\right)\right]$$

Capacitance per unit length, $C = \dfrac{2\pi\epsilon_0}{\left[\ln\left(\frac{b}{a}\right) + \frac{1}{\epsilon_r}\ln\left(\frac{c}{b}\right)\right]}$

Calculation of bound charges of co-axial cable system

$$\rho_b = -\vec{\nabla}\cdot\vec{P} \quad \text{and} \quad \sigma_b = \vec{P}\cdot\hat{n}$$

$$\vec{P} = \vec{D} - \epsilon_0\vec{E} \qquad \vec{D} = \frac{Q}{2\pi s l}(\hat{s}) \quad \text{and} \quad \vec{E} = \frac{Q\,(\hat{s})}{2\pi\epsilon_0 s l} \quad a < s < b$$

$$= \frac{Q\,(\hat{s})}{2\pi\epsilon s l} \quad b < s < c$$

For $a < s < b$:

$$\vec{P} = \vec{D} - \epsilon_0\frac{Q\,(\hat{s})}{2\pi\epsilon_0 s l} = \frac{Q\,\hat{s}}{2\pi s l} - \frac{Q\,\hat{s}}{2\pi s l} = 0$$

[which it should be as in vacuum there is no material to be polarized]

For $b < s < c$

$$\vec{P} = \vec{D} - \epsilon_0\vec{E} = \frac{Q\,(\hat{s})}{2\pi s l} - \epsilon_0\frac{Q\,(\hat{s})}{2\pi\epsilon s l} = \frac{Q\,(\hat{s})}{2\pi s l} - \frac{Q\,(\hat{s})}{2\pi\epsilon_r s l}$$

$$= \frac{Q}{2\pi s l}\left(1 - \frac{1}{\epsilon_r}\right)\hat{s} = \frac{Q}{2\pi s l}\left(\frac{\epsilon_r - 1}{\epsilon_r}\right)\hat{s}$$

$\therefore$ $\rho_b = -\vec{\nabla}\cdot\vec{P}$. Take divergence in cylindrical co-ordinates (only radial term)

$$\rho_b = -\frac{1}{s}\frac{\partial}{\partial s}(s P_s) = -\frac{1}{s}\frac{\partial}{\partial s}\left[\cancel{s}\cdot\frac{Q}{2\pi\cancel{s}l}\left(\frac{\epsilon_r - 1}{\epsilon_r}\right)\right] = 0$$

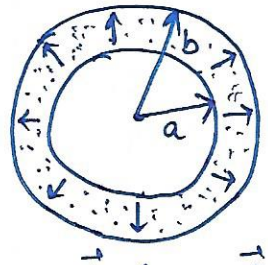
$$\sigma_b = \vec{P}\cdot\hat{n} = \vec{P}\cdot(-\hat{s})\Big|_{s=b} = -\frac{Q}{2\pi b l}\left(\frac{\epsilon_r - 1}{\epsilon_r}\right)$$

(for cylinder with radius b)

$$\sigma_b = \vec{P}\cdot\hat{n} = \vec{P}\cdot(\hat{s})\Big|_{s=c} = \frac{Q}{2\pi c l}\left(\frac{\epsilon_r - 1}{\epsilon_r}\right)$$

for cylinder with radius (c)

4. We have to find Electric field at $s<a$, $a<s<b$, $s>b$



$\vec{P} = \frac{k}{r}\hat{r}$

<u>First method:</u> (bound charges)

$$\sigma_b = \vec{P}\cdot\hat{n} = \vec{P}\cdot(-\hat{r})\Big|_{r=a} = -\frac{k}{r}\Big|_{r=a} = -\frac{k}{a}$$

[on the surface of sphere with radius 'a']

$$\sigma_b = \vec{P}\cdot\hat{n} = \vec{P}\cdot\hat{r} = \frac{k}{r}\hat{r}\cdot\hat{r}\Big|_{r=b} = \frac{k}{b}$$

[on the surface of sphere with radius 'b']

Clearly polarization vector points along the direction as shown in the figure.

$$\rho_b = -\vec{\nabla}\cdot\vec{P} = -\frac{1}{r^2}\frac{d}{dr}(r^2 P_r) = -\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{k}{r}\right) = -\frac{k}{r^2}$$

Now calculate fields due to σ_b and ρ_b due to Gauss's law

$$\epsilon_0\int\vec{E}\cdot d\vec{S} = Q_{total,enc} = Q_{f,enc} + Q_{b,enc}$$

But since there are no free charges in the problem, $Q_{f,enc} = 0$

i) <u>Region $r<a$</u>

$\boxed{\vec{E} = 0}$ as $Q_{b,enc} = 0$

ii) <u>Region $a<r<b$</u>

$$E\cdot 4\pi r^2 = \frac{1}{\epsilon_0}\left[\left(-\frac{k}{a}\right)\times 4\pi a^2 + \int_a^r \rho_b'(\vec{r}')\,dT'\right]$$

$$= \frac{1}{\epsilon_0}\left[-4\pi k a + \int_a^r -\frac{k}{r'^2}4\pi r'^2\,dr'\right] = \frac{1}{\epsilon_0}\left[-4\pi k a - 4\pi k\int_a^r dr'\right]$$

$$= \frac{1}{\epsilon_0}\left[-4\pi k r\right]$$

$$\therefore E\cdot 4\pi r^2 = -\frac{1}{\epsilon_0}\times 4\pi k r \Rightarrow \boxed{\vec{E} = -\frac{k}{\epsilon_0 r}\hat{r}}$$

iii) Region $r > b$

$$\oiint \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} Q_{enc,b}$$

Total bound charge enclosed within $r=a$ and $r=b$ is:

$$\int \sigma_b \, da + \int \rho_b \, d\tau = -\frac{k}{a} \times 4\pi a^2 + \frac{k}{b} \times 4\pi b^2 + \int_{r'=a}^{b} -\frac{k}{r'^2} \cdot 4\pi r'^2 dr$$

$$= -4\pi \cancel{k} a + 4\pi \cancel{k} b - 4\pi \cancel{k} b + 4\pi \cancel{k} a = 0$$

$$\cancel{= 4\pi k (b-r) - 4\pi k (r-b)}$$

$$\therefore \boxed{\vec{E} = \vec{0}}$$

Method 2

First let's calculate $\vec{D}$: $\oiint \vec{D} \cdot d\vec{a} = (Q_f)_{enc}$

$$= 0 \quad [\text{no free charges}]$$

Thus $\vec{D} = \vec{0} \Rightarrow \epsilon_0 \vec{E} + \vec{P} = \vec{0}$

$$\vec{E} = -\frac{\vec{P}}{\epsilon_0}$$

So electric field is only present where there is finite polarization $\vec{P}$ and is zero in other regions.

For $a < r < b$ $\quad \vec{E} = -\frac{\vec{P}}{\epsilon_0} = -\frac{k}{\epsilon_0 r} \hat{r}$ which agrees with

Method 1.

1. <u>Top plate</u>

metallic $\sigma_f = \sigma$

$\epsilon_r = 2$

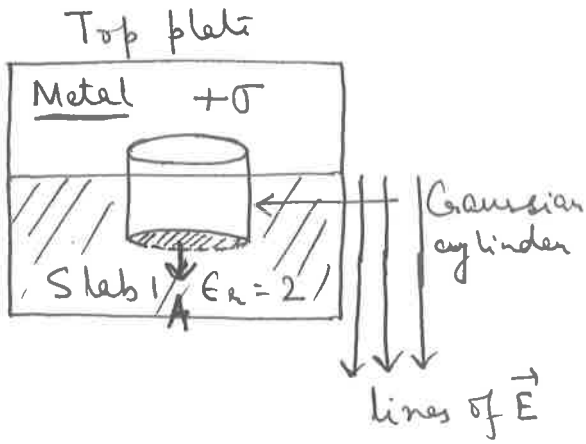
<u>Bottom plate</u>

$\sigma_f = -\sigma$

$\epsilon_r = 1.5$

(a) $D = ?$ (in each slab)

Apply $\oint \vec{D} \cdot d\vec{a} = (Q_f)_{enc}$ to Gaussian cylinder



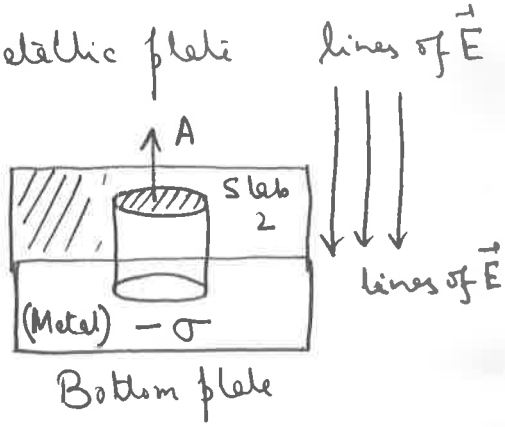
Inside metal (top plate) $\underline{D = 0}$

- so the gaussian cylinder has

$D \cdot A = +\sigma \times +A$ [contribution of A comes from circular base of cylinder embedded inside Slab 1]

$D = \sigma$

Apply the same to the bottom metallic plate



$\oint \vec{D} \cdot d\vec{a} = (Q_f)_{enc}$

$\therefore D \times (-A) = -\sigma A$ [inside metal $\underline{D = 0}$]

$\therefore D = \sigma$

[contribution of A comes from circular base of cylinder embedded inside Slab 2]

(b) $\vec{E}$ in each slab

~~Slab (a) $E = \frac{\sigma}{\epsilon_0 \epsilon_{r_1}} = \frac{\sigma}{2\epsilon_0}$~~

We know $(D)_{slab_1} = \sigma$ $\quad (D)_{slab_2} = \sigma$

$$\vec{D} = \epsilon \vec{E} \Rightarrow \vec{E} = \frac{\vec{D}}{\epsilon} = \frac{\vec{D}}{\epsilon_0 \epsilon_r}$$

$$E_1 = \frac{D}{\epsilon_0 \epsilon_{r_1}} = \frac{\sigma}{\epsilon_0 \times 2} = \frac{\sigma}{2\epsilon_0}$$

$$E_2 = \frac{D}{\epsilon_0 \epsilon_{r_2}} = \frac{\sigma}{\epsilon_0 \, 3/2} = \frac{2\sigma}{3\epsilon_0}$$

(c) Pol. in each slabs

$$\vec{P} = \vec{D} - \epsilon_0 \vec{E} \qquad P_1 = D_1 - \epsilon_0 E_1$$

$$= \sigma - \epsilon_0 \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{2}$$

$$P_2 = D_2 - \epsilon_0 E_2$$

$$= \sigma - \epsilon_0 \frac{2\sigma}{3\epsilon_0} = \frac{\sigma}{3}$$

(d) Voltage difference between the plates

$$V = E_1 a + E_2 a$$

$$= \frac{\sigma}{2\epsilon_0} a + \frac{2\sigma}{3\epsilon_0} a = \frac{7\sigma a}{6\epsilon_0}$$

(e) Polarizations in each slab are uniform, $\rho_b = -\vec{\nabla} \cdot \vec{P} = 0$

$\sigma_b = \vec{P} \cdot \hat{n}$ [take $\hat{n}$ = +ve for each outward pointing area normal) for each metal plate

Slab 1

Top part

$$\sigma_b = P_1 \cdot (+\hat{n})$$

$$= \frac{-\sigma}{2}$$

Bottom part

$$\sigma_b = P_1 \cdot (+\hat{n})$$

$$= +\sigma/2$$

Slab 2

Top part

$$\sigma_b = P_2 \cdot (+\hat{n})$$

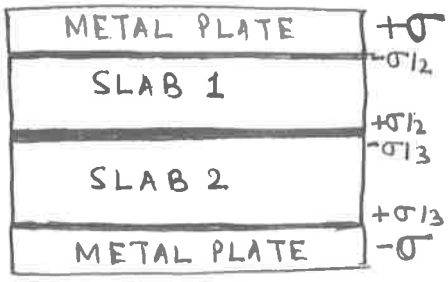
$$= \frac{\sigma}{3}$$

Bottom part

$$\sigma_b = P_2 (-\hat{n})$$

$$= \frac{-\sigma}{3}$$

(e) Field inside slab 1



Total surface charge on top plate $= \sigma - \frac{\sigma}{2} = \frac{\sigma}{2}$

Total surface charge on bottom plate, $\frac{\sigma}{2} - \cancel{\frac{\sigma}{3}} + \cancel{\frac{\sigma}{3}} - \sigma$

$= -\frac{\sigma}{2}$

Field inside slabs 1 $= \frac{\sigma}{2\epsilon_0}$

$\boxed{\text{Field due to surface charge } \sigma = \sigma/\epsilon_0}$

Field inside slab 2

Total surface charge on top plate

$$\sigma - \frac{\sigma}{2} + \frac{\sigma}{2} - \frac{\sigma}{3} = \frac{2\sigma}{3}$$

Total surface charge on bottom plate

$$\frac{\sigma}{3} - \sigma = -\frac{2\sigma}{3}$$

$\therefore$ Field inside slab 2

$$= \frac{2\sigma}{3\epsilon_0}.$$